

ICT4Ag | AI & DATA-DRIVEN SOLUTIONS FOR SUSTAINABILITY

PROFILE SUMMARY

Highly motivated and results-oriented professional with 10+ years of experience specialising in the intersection of Agricultural Engineering, Data Science, and ICT4D. Proven ability to conceptualise, develop, and deploy technology-backed solutions from AI/ML predictive models for food systems to water systems for climate resilience. Extensive experience leading multi-disciplinary projects across Southern Africa funded by major donors (USAID, World Bank, GCF, FAO). Seeking consulting roles where we can transform complex data into actionable, sustainable solutions.

CORE COMPETENCIES & TECHNICAL SKILLS

Category	Expertise & Tools
Data Science & AI/ML	Data Collection & Configuration (Kobo Toolbox, CSPRO), AI Development (CTO Level), Machine Learning (Predictive Models), Data Modelling, Python, R, SPSS, SQL, APIs, Git, databases, systems design.
Engineering design and reporting	Water development & Feasibility Studies (CropWat, DSSAT, AutoCAD), Water Management (MUS, Groundwater Assessment), Climate Profiling, Field Infrastructure Mapping, Postharvest technology, production systems, GIS Analysis (QGIS, ILWIS)
Project & Consultancy Mgt	Project Lifecycle Management (Conception to Completion), Budget Control, Proposal Development & Negotiation, Monitoring & Evaluation (M&E), MS Office
Capacity Building	Training Program Design and Delivery (Water Management, Irrigation, Data Management) for farmers, extensionists, and trainers (e.g., FAO, USAID).

PROFESSIONAL EXPERIENCE

CHIEF TECHNOLOGY OFFICER (CTO)

TINYA Pvt Ltd | Harare, Zimbabwe | 2023 – Present

- Leads technology development, AI/ML implementation, and software architecture for client-specific business and agricultural solutions.
- Achievement: Placed 3rd in the 2024 POTRAZ Innovation Drive Hackathon.

CONSULTANCY PORTFOLIO (2018 – Present)

Project/Client	Role & Focus Area	Key Deliverables & Outcomes
Green Climate Fund (GCF) Project (Sesani Projects / Chipinge District)	Engineering Consultant	Conducted detailed crop water requirement calculations, value chain analysis, and delivered comprehensive irrigation design and agribusiness reports for five critical irrigation schemes.
Seke and Pfura District Master Plans	Agricultural Consultant	Led field data collection, Key Informant Interviews (KIIs), and water use analysis to prepare detailed proposals for technology-backed agricultural systems, irrigation infrastructure, and value chains.

Smart Village Development (SAGIT)	Lead ICT & Agricultural Engineering Consultant	Developed the initial concept document and preliminary investigation methodology for the ICT4Ag components of a pan-African smart village architectural project.
Successful Partnerships for MUS (USAID/Save The Children)	Research Support & Technical Backstopping	Supervised evaluation, qualitative/quantitative data collection, analysis, and reporting across 20 Multiple-Use Water Services (MUS) projects in six districts.
SIRP Smallholder Irrigation Revitalisation Programme (ARUP Zimbabwe)	Engineering Consultant / Technical Lead	Led climate profiling and developed climate-proofed crop water requirements for 30 irrigation schemes. Led feasibility studies, field infrastructure mapping, and preliminary design for over 7 irrigation schemes.
Diamondback Moth (DBM) Management System (BIUST – Botswana)	Lead ICT Consultant	Developed the backend machine learning model and frontend Progressive Web App (PWA) used as a decision support tool for managing the DBM pest.

(Detailed work history available upon request.)

RESEARCH & PROJECT MANAGEMENT (Pre-2018)

Assistant Project Manager – CLIRCS Project

University of Zimbabwe / FAO / EU | 09/2013 – 01/2018

- Managed field research activities for the CLIRCS project across three districts (Mbire, Guruve, Hwedza).
- Responsibilities included research design, smallholder community profiling, participatory data collection & management, M&E, and publication.
- Innovation: Developed a prototype machine learning-based decision support tool for larger grain borer threat prediction.

EDUCATION & PROFESSIONAL DEVELOPMENT

DATES	QUALIFICATION, INSTITUTION
2021	MPhil (ICT And Postharvest Science), Department of Agricultural and Biosystems Engineering, University of Zimbabwe
2010	BSc. Honours in Agricultural Engineering, Department of Agricultural and Biosystems Engineering, University of Zimbabwe
2003	Basic Workshop Practice Certificate, Windhoek Vocational Training College, Windhoek, Namibia
2004	Civil Engineering, University of Namibia; Civil Engineering, University of Cape Town (Incomplete)

RELEVANT TRAINING & CERTIFICATION

- ILO Start Your Business (SYB) Certificate, 2025
- Introduction to Ethics in Evidence Generation (Basic) (UNICEF), 2022
- Digital Innovation Profile Co-creation Workshop (International Telecommunications Union, 2023)
- FAO Training of Trainers (Food Loss Analysis Methodology, 2018)
- Overview of the Integrated Food Security Phase Classification (IPC) (2023)
- The Complete Machine Learning Course with Python (Udemy, 2020)
- Adjudicator for POTRAZ Innovation Hackathons (2021 & 2022)
- International Computer Driving License (ICDL), 2006

SELECTED PUBLICATIONS

- Manzungu E., Cobbing J., Troell J., Van Koppen B., Mapedza E., Mwakiwa E., Nhwatiwa T., **Nyabako T.** Takawira D., Nkomo L., and Mayer J., 2024. Financial sustainability of multiple use water systems: experiences from Zimbabwe, and implementation and policy implications for sub-Saharan Africa. *Water SA* (Under review).
- **Nyabako T.**, et al. (2024). Using climate variables to predict flight behaviour of the larger grain borer, *Prostephanus truncatus* (Horn)... (*JSPR*, Under review).
- **Nyabako T.**, et al. (2020). Predicting *Prostephanus truncatus* (Horn) populations and insect grain damage in smallholder farmers' maize stores: **A machine learning approach**. *Journal of Stored Product Research*. (DOI: 10.1016/j.jspr.2020.101592).
- **Nyabako T.** and Manzungu E. (2012). An Assessment of the Adaptability to Climate Change of Commercially Available Maize Varieties in Zimbabwe. *Environment and Natural Resources Research*.
- **Nyabako T.** and Manzungu E. (2011). Development of a GPRS Internet Based Data Management System for Effective Dam Operation in Zimbabwe. *International Journal of Business, Humanities and Technology*.

(Full list of 14 publications available upon request.)

REGIONAL EXPERIENCE & LANGUAGES

Area	Detail
Geographical Experience	Zimbabwe (Extensive), South Africa, Zambia, Namibia, Botswana.
Languages	English & Shona (Native), Nyanja (Limited Proficiency), French (Elementary).

REFERENCES

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(Additional references are available upon request.)